Wave

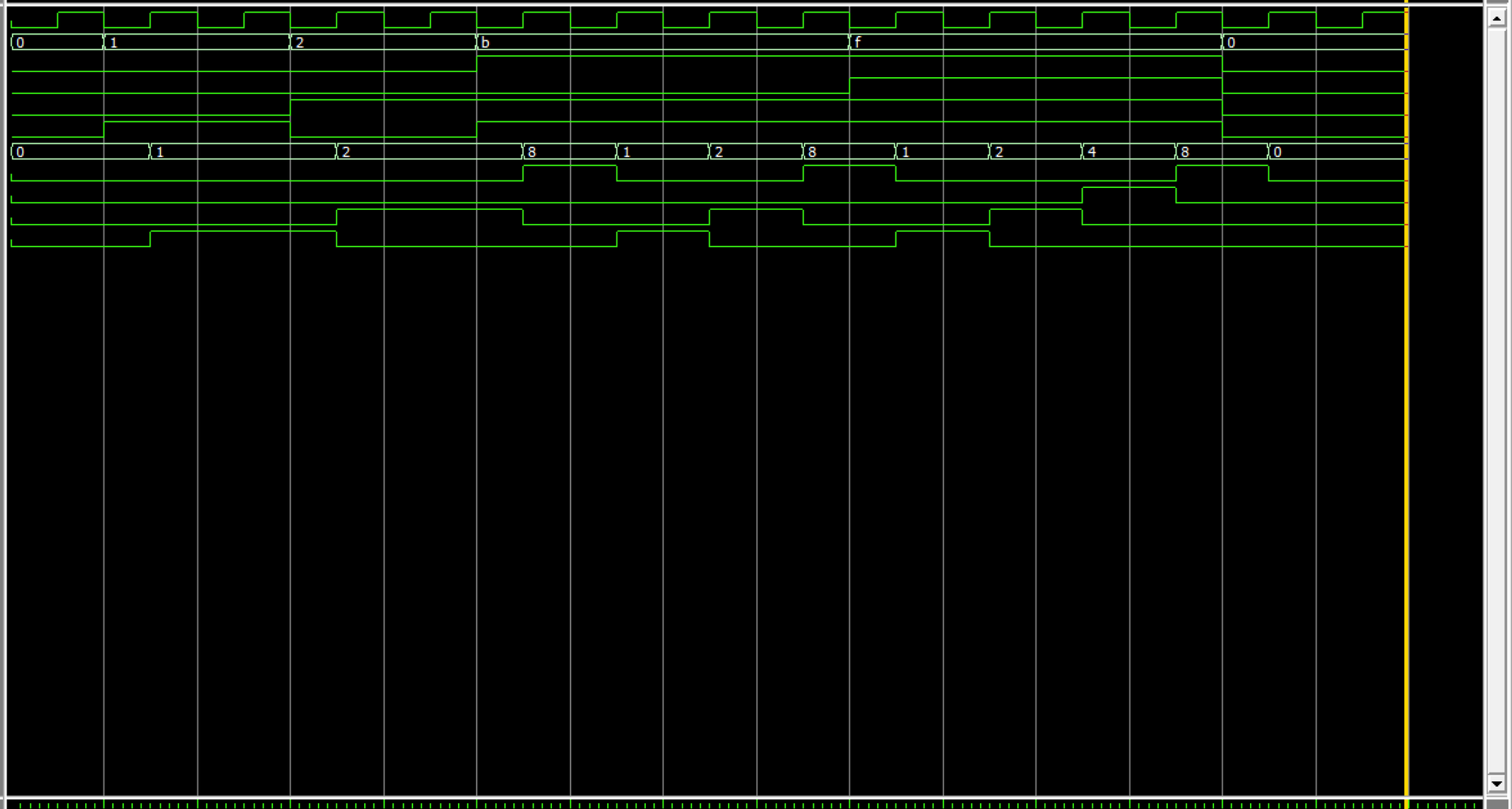
File Edit View Add Format Tools Bookmarks Window Help

Wave - Default

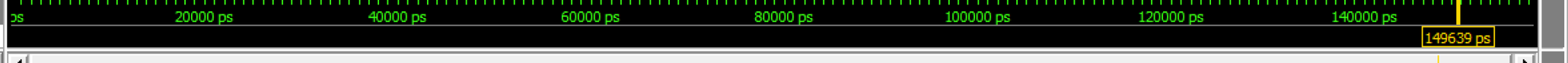
100 ps

Search:

Msgs	
/RRA_tb/clk	1'h1
/RRA_tb/req	4'h0
[3]	0
[2]	0
[1]	0
[0]	0
/RRA_tb/gnt	4'h0
[3]	1'h0
[2]	1'h0
[1]	1'h0
[0]	1'h0



Now	150000 ps
Cursor 1	149639 ps



0 ps to 157500 ps

/RRA_tb/gnt

Time	req[3]	req[2]	req[1]	req[0]	gnt[3]	gnt[2]	gnt[1]	gnt[0]
20ns	0	0	0	1	0	0	0	1
40ns	0	0	1	0	0	0	1	0
50ns	1	0	1	1	0	0	1	0
55ns	1	0	1	1	1	0	0	0
65ns	1	0	1	1	0	0	0	1
75ns	1	0	1	1	0	0	1	0
85ns	1	0	1	1	1	0	0	0
90ns	1	1	1	1	1	0	0	0
95ns	1	1	1	1	0	0	0	1
105ns	1	1	1	1	0	0	1	0
115ns	1	1	1	1	0	1	0	0
125ns	1	1	1	1	1	0	0	0
135ns	0	0	0	0	0	0	0	0

The state table of the testbench is given above, and it shows that the grant is given to only one requestor at a given clock cycle and it rotates through the requestors. The priority of the arbitrer changes every posedge of the clock regardless of whether the previous requestor has withdrawn his request or not. This is because if one requestor requests access for an indefinite amount of time, fair chance of grant should be given to the other requestors as well.